

HRMS – Leave Request

Professional Functional, Technical & Non-Functional Test Specification

Purpose: A complete QA test specification for the Leave Request module, covering business functionality, API/backend/database behavior, security, performance, reliability, concurrency, maintainability and regression testing.

Base URL: http://localhost:5168/

Testing principle: A feature is considered tested only when API behavior, business rules, calculated values and database state agree with the expected result.

1. Test Scope

The scope covers the complete Leave Request lifecycle: Create → DRAFT → Update → Submit → PENDING → Approve/Reject, plus Cancel and Delete where permitted. It also covers full-day and half-day calculations, Company Holiday deduction, validation, overlap prevention, status transitions, pagination, persistence, error handling, security, concurrency, performance and regression.

2. Test Environment & Reference Data

Item	Value
CompanyId	023FF1F8-4DD0-4942-1955-08DEFD16782B
FY2027 LeaveYearId	8BB8836A-3920-4DA2-41CF-08DEFD305CC8
EmployeeId	19ED1DF1-C3CC-4614-85F0-08DEF7B598A7
LeaveTypeId	9A5FF5C9-49DF-43F4-5107-08DEFD337F69
Full Day Part	30000000-0000-0000-0000-000000000001
Half Day Part	30000000-0000-0000-0000-000000000002
Known holiday test dates	2027-04-23 and 2027-04-30

3. Test Categories

Functional: verifies what the business feature does.

Technical: verifies API contracts, handlers, business rules, repository behavior, calculations, persistence, transactions and concurrency.

Non-functional: verifies performance, security, reliability, scalability, logging, maintainability and resilience.

4. Priority Model

P0 – Critical: failure blocks release. Includes create, submit, approval workflow, TotalDays, holiday calculation, overlap, status transitions, authorization and database consistency.

P1 – High: important feature behavior such as update, cancel, delete, half-day, multiple holidays, pagination and validation.

P2 – Medium: logging, error wording, large-input validation and uncommon edge cases.

P3 – Low: non-critical metadata or presentation details.

5. Functional Test Cases

ID	Scenario	Priority	Expected Result
FL-LR-001	Create normal full-day leave	P0	Request created as DRAFT; correct TotalDays
FL-LR-002	Invalid EmployeeId	P0	Request rejected; no invalid record
FL-LR-003	Employee does not exist	P0	Request rejected
FL-LR-004	Invalid LeaveYearId	P0	Request rejected
FL-LR-005	Invalid LeaveTypeId	P0	Request rejected
FL-LR-006	Invalid StartDayPartId	P1	Request rejected

ID	Scenario	Priority	Expected Result
FL-LR-007	Invalid EndDayPartId	P1	Request rejected
FL-LR-008	Missing EmployeeId	P1	400 validation
FL-LR-009	Missing LeaveYearId	P1	400 validation
FL-LR-010	Missing LeaveTypeId	P1	400 validation
FL-LR-011	Missing FromDate	P1	400 validation
FL-LR-012	Missing ToDate	P1	400 validation
FL-LR-013	Empty reason	P1	Validation according to configured rule
FL-LR-014	Invalid GUID	P1	400 validation

6. Date Validation Tests

FL-DATE-001 — FromDate = ToDate. Expected: one valid day for a normal full-day request. FL-DATE-002 — FromDate later than ToDate. Expected: rejected. FL-DATE-003 — Request outside FY2027 (2027-04-01 to 2028-03-31). Expected: rejected according to LeaveYear validation. FL-DATE-004 — Boundary start date 2027-04-01. Expected: accepted if other rules pass. FL-DATE-005 — Boundary end date 2028-03-31. Expected: accepted if other rules pass. FL-DATE-006 — Date outside selected LeaveYear. Expected: rejected.

7. Full-Day Calculation Tests

FL-CAL-001 — 2027-05-05 → 2027-05-05 = 1.00. FL-CAL-002 — 2027-05-05 → 2027-05-07 = 3.00 when no eligible holidays exist. FL-CAL-003 — 2027-05-05 → 2027-05-10 = 6.00 when no eligible holidays exist.

8. Half-Day Calculation Tests

FL-CAL-004 — One normal half-day request = 0.50. FL-CAL-005 — Multi-date request with valid start/end day-part combination = expected value according to configured DayPart rules. FL-CAL-006 — Invalid start/end day-part combination = rejected.

9. Company Holiday Calculation — Critical

The current calculation considers a CompanyHoliday when: • LeaveYearId matches the request. • HolidayDate is within FromDate and ToDate. • IsDeleted = false. • IsActive = true. • IsOptional = false.

FL-HOL-001 — Holiday in middle. Leave: 2027-04-21 → 2027-04-25. Holiday: 2027-04-23. Expected: 21=1, 22=1, 23=0, 24=1, 25=1 → TotalDays=4.00.

This scenario has already been successfully tested in the project.

FL-HOL-002 — Holiday on first date. Holiday 2027-05-10; leave 2027-05-10 → 2027-05-12. Expected full-day result: 2.00.

FL-HOL-003 — Holiday on last date. Holiday 2027-05-12; leave 2027-05-10 → 2027-05-12. Expected: 2.00. FL-HOL-004 — Single-day full-day holiday. Holiday and leave both 2027-05-10. Expected: 0.00. FL-HOL-005 — Multiple holidays. Leave 2027-05-10 → 2027-05-14; holidays May 10, 12, 14. Expected: 2.00.

10. Holiday Status Tests

ID	Holiday Condition	Expected
FL-HOL-006	Active + non-optional + not deleted	Deduct
FL-HOL-007	Optional holiday	Do not deduct
FL-HOL-008	Inactive holiday	Do not deduct
FL-HOL-009	Deleted holiday	Do not deduct
FL-HOL-010	Holiday belongs to another LeaveYear	Do not deduct

11. Create → Update Tests

FL-UPD-001 — Create a DRAFT for 2027-05-05 → 2027-05-07. Update it to 2027-05-05 → 2027-05-10. Expected: update succeeds and TotalDays is recalculated. FL-UPD-002 — Update a PENDING request. Expected HTTP 409 with: “Only draft leave requests can be updated.” FL-UPD-003 — Update a DRAFT from 2027-04-21 → 2027-04-25 while an eligible holiday exists. Expected TotalDays=4.00. FL-UPD-004 — Route ID differs from request ID. Expected request rejected; no other request may be modified.

12. Submit Tests

FL-SUB-001 — DRAFT → Submit. Expected DRAFT changes to the submission/pending state according to the current status master. FL-SUB-002 — Submit recalculation. Create 2027-05-10 → 2027-05-12 when no holiday exists (3.00), add holiday 2027-05-11, then submit. Expected final calculation 2.00. FL-SUB-003 — Submit an invalid/non-DRAFT request. Expected rejection according to current status rules. FL-SUB-004 — Submit the same request twice. Expected no duplicate workflow and no invalid second transition.

Verified submit endpoint example: POST /api/leave-requests/DAAA731C-86A6-4F2F-1EB4-08DEFF48E482/submit

13. Overlap Tests — Critical

FL-OVR-001 — Existing PENDING leave 2027-04-21 → 2027-04-25; new request 2027-04-22 → 2027-04-24. Expected 409. FL-OVR-002 — Same start date. Expected rejection. FL-OVR-003 — Same end date. Expected rejection. FL-OVR-004 — New request completely contains existing request. Expected rejection. FL-OVR-005 — New request is immediately after existing request with no date overlap. Expected allowed if all other rules pass.

Verified error example: “Employee already has a pending or approved leave request for the selected dates.”

14. Approval Tests

FL-APP-001 — PENDING → APPROVED. Expected successful transition. FL-APP-002 — Approve DRAFT. Expected rejection. FL-APP-003 — Approve REJECTED. Expected rejection. FL-APP-004 — Approve CANCELLED. Expected rejection. FL-APP-005 — Verify database StatusId after approval.

15. Rejection Tests

FL-REJ-001 — PENDING → REJECTED. Expected successful transition. FL-REJ-002 — Reject DRAFT. Expected rejection. FL-REJ-003 — Reject APPROVED. Expected rejection. FL-REJ-004 — Verify rejection information/audit data if supported by the current model.

16. Cancellation Tests

FL-CAN-001 — Cancel a request from an allowed source state. Expected CANCELLED. FL-CAN-002 — Cancel an already CANCELLED request. Expected rejection. FL-CAN-003 — Verify database status and audit information.

17. Delete Tests

FL-DEL-001 — Delete an allowed request. Expected soft-delete behavior if configured. FL-DEL-002 — Verify IsDeleted=1. FL-DEL-003 — Deleted request must not appear in normal active queries. FL-DEL-004 — Deleted request must not participate in CompanyHoliday or LeaveRequest business calculations where the current soft-delete rules exclude it.

18. Technical/API Test Cases

ID	Technical Scenario	Expected
API-001	Wrong HTTP method	405 or configured routing response
API-002	Invalid Content-Type	400/content-negotiation error
API-003	Malformed JSON	400 Bad Request
API-004	Unknown request GUID	404/not found
API-005	Invalid GUID format	400 validation
API-006	Route/body IDs differ	400/409; no wrong record update
API-007	Unexpected server exception	Standard 500 response; no internals exposed

ID	Technical Scenario	Expected
API-008	Pagination page 1 size 10	Correct page metadata and items
API-009	Large page number	Correct empty/valid page behavior
API-010	Cancellation token/request cancellation	No inconsistent persistence

19. Handler / Business Rules Technical Tests

TECH-BR-001 — Verify Create calls employee validation. TECH-BR-002 — Verify LeaveYear validation occurs before date calculation. TECH-BR-003 — Verify LeaveType validation. TECH-BR-004 — Verify both day parts are validated. TECH-BR-005 — Verify date validation. TECH-BR-006 — Verify day-part combination validation. TECH-BR-007 — Verify overlap check. TECH-BR-008 — Verify Create uses CalculateTotalDaysAsync. TECH-BR-009 — Verify Submit recalculates using LeaveYearId. TECH-BR-010 — Verify Update recalculates using LeaveYearId. TECH-BR-011 — Verify CompanyHoliday repository filter is applied correctly. TECH-BR-012 — Verify business logic is not duplicated in controllers.

20. Database & Persistence Tests

DB-001 — After Create, verify EmployeeId, LeaveYearId, LeaveTypeId, day parts, dates, TotalDays, StatusId and Reason. DB-002 — After Update, verify changed fields, TotalDays and audit fields. DB-003 — After Submit, verify status and appropriate timestamps. DB-004 — After approval/rejection/cancellation, verify StatusId. DB-005 — Verify soft-delete fields after Delete. DB-006 — Verify no duplicate records are created by repeated requests. DB-007 — Verify foreign-key relationships. DB-008 — Verify CompanyHoliday records used for calculation belong to the selected LeaveYear.

21. SQL Verification Queries

```
SELECT Id, EmployeeId, LeaveYearId, LeaveTypeId, StartDayPartId, EndDayPartId, StatusId, FromDate, ToDate, TotalDays, Reason, AppliedOn FROM LeaveRequests WHERE Id = '';
```

```
SELECT Id, LeaveYearId, HolidayDate, Name, IsOptional, IsActive, IsDeleted FROM CompanyHolidays WHERE LeaveYearId = '8BB8836A-3920-4DA2-41CF-08DEFD305CC8' ORDER BY HolidayDate;
```

22. Transaction & Consistency Tests

TECH-TXN-001 — Force database failure during Create. Expected: no partially persisted leave request. TECH-TXN-002 — Force failure during Submit. Expected: request does not remain in an invalid intermediate state. TECH-TXN-003 — Failure after status change but before persistence must not expose an inconsistent workflow state.

23. Concurrency Tests — Critical

TECH-CON-001 — Open two browser/API clients for the same employee and submit overlapping requests simultaneously. Expected: only the permitted request succeeds; the system must prevent a race-condition duplicate. TECH-CON-002 — Two users update the same DRAFT simultaneously. Verify defined last-write/concurrency behavior. TECH-CON-003 — Two approvers attempt to approve the same PENDING request simultaneously. Only one valid transition should occur.

24. Security / Authorization Tests

SEC-001 — Employee attempts to access another employee's leave request. Expected 401/403 or appropriate authorization response. SEC-002 — IDOR test: change request GUID in URL. Unauthorized data must not be exposed or modified. SEC-003 — Employee attempts to approve a request without permission. Expected authorization failure. SEC-004 — Employee attempts to modify a request belonging to another employee. Expected authorization failure. SEC-005 — SQL injection strings in searchable/filterable input. Expected no injection. SEC-006 — Malicious/oversized Reason input. Expected safe validation and no database compromise. SEC-007 — Error responses must not expose connection strings, SQL, stack traces or secrets.

25. Non-Functional Testing

Non-functional testing answers: "How well does the Leave Request feature work under real-world conditions?"

26. Performance Tests

ID	Scenario	Target / Expected
NFR-PERF-001	Normal Create	Prefer <500 ms; confirm project SLA
NFR-PERF-002	Normal Get/List	Prefer <500 ms; confirm project SLA
NFR-PERF-003	Update	Prefer <500 ms; confirm project SLA
NFR-PERF-004	Submit	Prefer <500 ms–1 s; confirm project SLA
NFR-PERF-005	Large date range	Correct result without excessive CPU/DB load
NFR-PERF-006	100+ CompanyHolidays	Correct result with acceptable response time
NFR-PERF-007	100,000+ leave records	Pagination remains stable
NFR-PERF-008	Concurrent requests	No unacceptable degradation or incorrect data

27. Scalability Tests

NFR-SCL-001 — Large LeaveRequest history. Verify queries remain paginated. NFR-SCL-002 — Large CompanyHoliday table across many LeaveYears. Verify holiday lookup remains scoped by LeaveYear and date range. NFR-SCL-003 — High concurrent submission volume. Verify database and business-rule consistency. NFR-SCL-004 — Growth test with realistic production-sized data set.

28. Reliability & Resilience Tests

NFR-REL-001 — Repeat Create → Submit → Approve flow many times. No random workflow failure. NFR-REL-002 — Retry an API call after timeout. Verify no unintended duplicate. NFR-REL-003 — Database temporarily unavailable. API returns controlled error and recovers after database recovery. NFR-REL-004 — Submit same request twice. Second call must not create a duplicate workflow. NFR-REL-005 — Cancel HTTP request while server is processing. Verify database consistency.

29. Logging & Observability

NFR-LOG-001 — Business 409 errors contain useful TraceId. NFR-LOG-002 — Unexpected exceptions are logged server-side. NFR-LOG-003 — Logs contain operation/request context sufficient for troubleshooting. NFR-LOG-004 — Sensitive personal information is not unnecessarily logged. NFR-LOG-005 — TraceId from response can be correlated with server logs.

30. Maintainability / Architecture Tests

NFR-MNT-001 — No duplicate TotalDays calculation exists in Create, Submit and Update. NFR-MNT-002 — Holiday logic remains in business-rule/application logic rather than controllers. NFR-MNT-003 — Repository abstraction is respected. NFR-MNT-004 — Handlers remain orchestration-focused. NFR-MNT-005 — Changes to CompanyHoliday logic do not break unrelated LeaveRequest behavior. NFR-MNT-006 — Build succeeds with zero compile errors and warnings are reviewed.

31. Regression Test Suite

After every change to LeaveRequest or CompanyHoliday calculation, rerun at minimum: Create normal leave; Create half-day; invalid employee; invalid LeaveYear; invalid LeaveType; invalid dates; overlap; holiday deduction; Update DRAFT; Update PENDING; Update with holiday; Submit DRAFT; Submit recalculation; invalid Submit; Approve PENDING; invalid Approve; Reject PENDING; invalid Reject; Cancel; Delete/soft delete; Get/List; pagination; database TotalDays; database StatusId; authorization; duplicate Submit; concurrency.

32. Known Project Scenarios Already Tested

A6562C90-4F07-49D9-B1AF-08DEFED676EE — 2027-04-21 → 2027-04-25. Baseline 5.00; holiday-aware test resulted in 4.00. DAAA731C-86A6-4F2F-1EB4-08DEFF48E482 — 2027-04-21 → 2027-04-25. Holiday-aware result 4.00.

A6562C90-4F07-49D9-B1AF-08DEFED676EE — later update test around 2027-04-28 → 2027-05-02 resulted in 4.00 with the applicable holiday. Verified 409: “Only draft leave requests can be updated.” Verified 409: “Employee already has a pending or approved leave request for the selected dates.”

33. Sample Test Execution Sheet

Test ID	Category	Priority	Expected	Actual	Status
FL-LR-001	Functional	P0	DRAFT + correct TotalDays		■
FL-HOL-001	Functional	P0	4.00		■
FL-HOL-002	Functional	P0	2.00		■
FL-HOL-003	Functional	P0	2.00		■
FL-HOL-004	Functional	P0	0.00		■
FL-HOL-005	Functional	P0	2.00		■
FL-UPD-002	Functional	P0	409		■
FL-SUB-001	Functional	P0	PENDING		■
FL-OVR-001	Functional	P0	409		■
FL-APP-001	Functional	P0	APPROVED		■
FL-REJ-001	Functional	P0	REJECTED		■
FL-CAN-001	Functional	P1	CANCELLED		■
FL-DEL-001	Functional	P1	Soft deleted		■
API-003	Technical	P1	400		■
DB-001	Technical	P0	Correct DB row		■
TECH-CON-001	Technical	P0	No race-condition duplicate		■
SEC-002	Security	P0	IDOR blocked		■
NFR-PERF-001	Performance	P1	Within SLA		■
NFR-REL-004	Reliability	P0	Safe retry		■
NFR-MNT-001	Maintainability	P1	No duplicated calculation		■

34. Entry Criteria

Testing can start when: • Application builds successfully. • Database is available. • Required LeaveYear, Employee, LeaveType, DayParts and CompanyHoliday data exist. • API is running. • Test environment is isolated from production. • QA has the current API/command contract.

35. Exit Criteria

Leave Request should be considered QA-complete only when: • All P0 functional tests pass. • P0/P1 technical tests pass. • Security-critical tests pass. • Defined performance targets pass. • Regression suite passes. • No open Critical/High defects remain. • Database integrity is confirmed. • Calculation scenarios are verified. • Evidence is recorded for failed/retested cases.

36. Defect Severity Guidance

Critical: data corruption, unauthorized access, duplicate approved leave, incorrect TotalDays affecting payroll/leave balance, or inability to use the workflow. **High:** Create/Update/Submit/Approve/Reject failure, incorrect holiday deduction, broken overlap rule. **Medium:** non-critical validation or response formatting issue. **Low:** cosmetic or minor wording issue.

37. Final QA Recommendation

The Leave Request module should be tested as a complete workflow rather than as isolated CRUD endpoints. The most important release gates are TotalDays correctness, CompanyHoliday behavior, overlap prevention, status transitions, authorization, database consistency and concurrency. Any failure in these areas should block release until investigated and regression-tested.

End of Professional QA Test Specification